

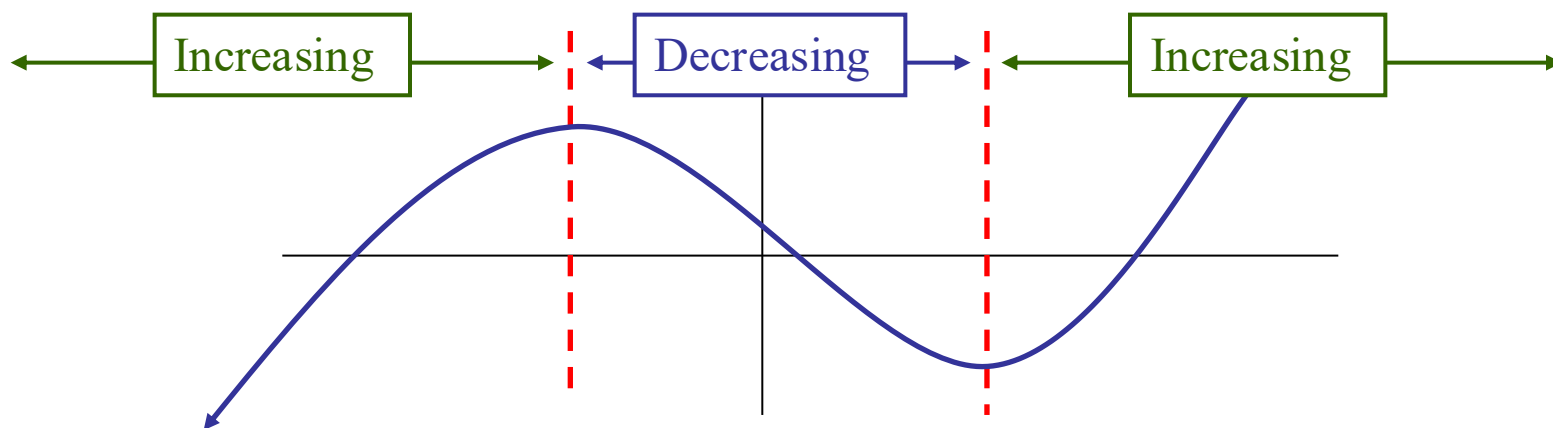
Applications of the Derivative

- Applications of the First Derivative
- Applications of the Second Derivative
- Curve Sketching
- Optimization I
- Optimization II

Increasing/Decreasing

A function f is **increasing** on (a, b) if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.

A function f is **decreasing** on (a, b) if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$.



Increasing/Decreasing/Constant



If $f'(x) > 0$ for each value of x in an interval (a, b) , then f is increasing on (a, b) .



If $f'(x) < 0$ for each value of x in an interval (a, b) , then f is decreasing on (a, b) .



If $f'(x) = 0$ for each value of x in an interval (a, b) , then f is constant on (a, b) .

Sign Diagram to Determine where $f(x)$ is Inc./Dec.

Steps:

1. Find all values of x for which $f'(x) = 0$ or $f'(x)$ is discontinuous and identify open intervals with these points.
2. Test a point c in each interval to check the sign of $f'(c)$.
 - a. If $f'(c) > 0$, f is *increasing* on that interval.
 - b. If $f'(c) < 0$, f is *decreasing* on that interval.

Example

Determine the intervals where $f(x) = x^3 - 6x^2 + 1$ is increasing and where it is decreasing.

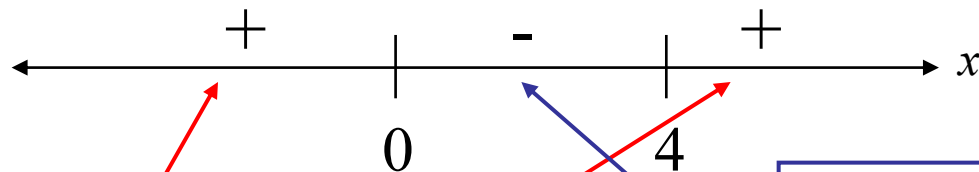
$$f'(x) = 3x^2 - 12x$$

$$3x^2 - 12x = 0$$

$$3x(x - 4) = 0$$

$$3x = 0 \text{ or } x - 4 = 0$$

$$x = 0, 4$$



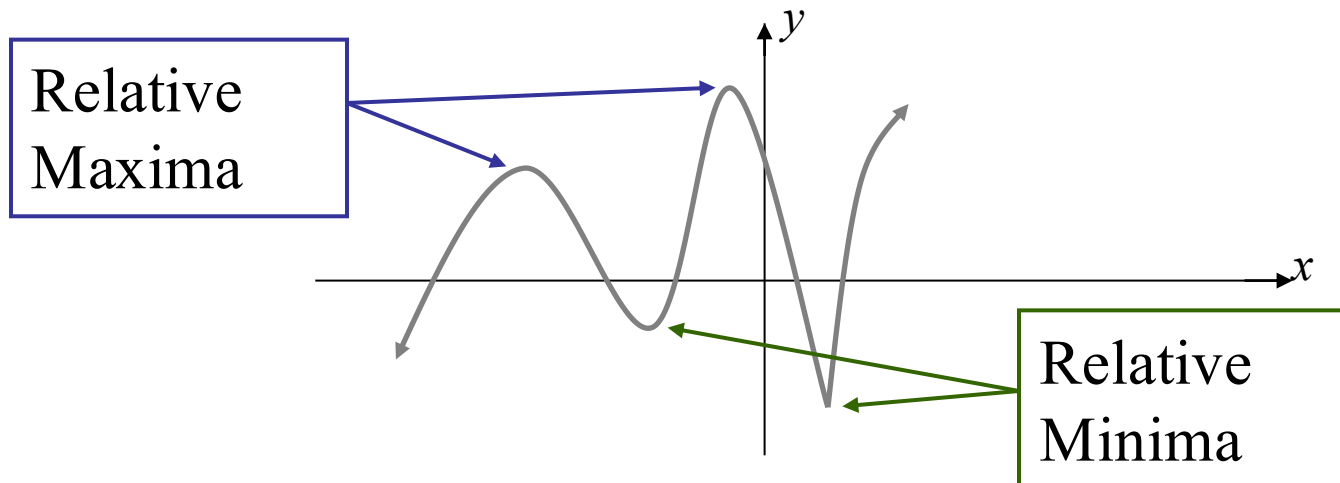
f is increasing
on $(-\infty, 0) \cup (4, \infty)$

f is decreasing
on $(0, 4)$

Relative Extrema

A function f has a relative maximum at $x = c$ if there exists an open interval (a, b) containing c such that $f(x) \leq f(c)$ for all x in (a, b) .

A function f has a relative minimum at $x = c$ if there exists an open interval (a, b) containing c such that $f(x) \geq f(c)$ for all x in (a, b) .

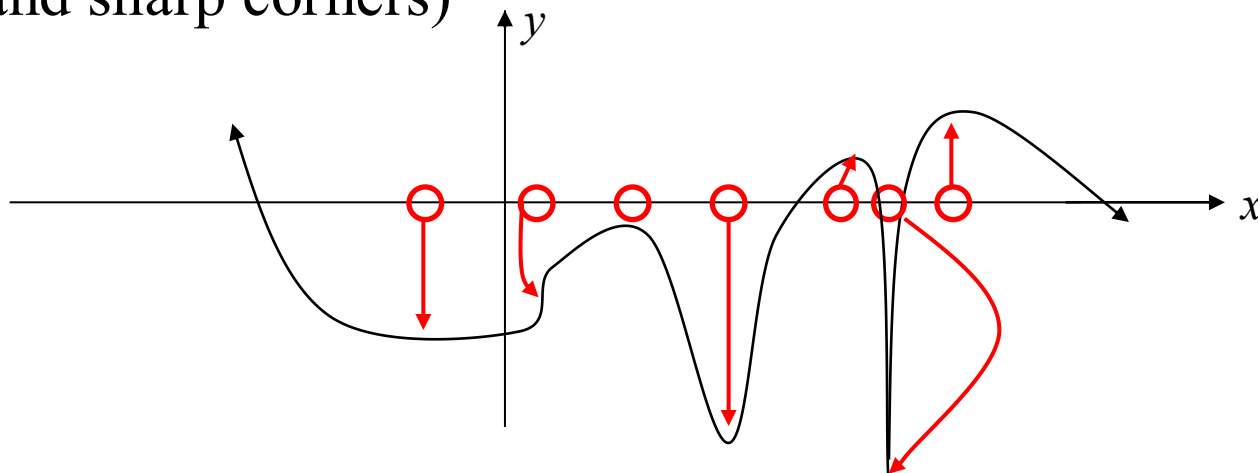


Critical Numbers of f

A ***critical number*** of a function f is a number in the domain of f where

$$f'(x) = 0 \text{ or } f'(x) \text{ does not exist.}$$

(horizontal tangent lines, vertical tangent lines and sharp corners)



The First Derivative Test

1. Determine the critical numbers of f .
2. Determine the sign of the derivative of f to the left and right of the critical number.

left	right	
+	−	$f(c)$ is a relative maximum
−	+	$f(c)$ is a relative minimum
No change		No relative extremum

Example

Find all the relative extrema of $f(x) = x^3 - 6x^2 + 1$.

$$f'(x) = 3x^2 - 12x$$

$$3x^2 - 12x = 0$$

$$3x(x - 4) = 0$$

$$3x = 0 \text{ or } x - 4 = 0$$

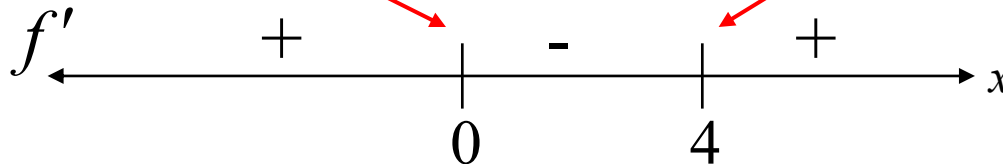
$$x = 0, 4$$

Relative max.

$$f(0) = 1$$

Relative min.

$$f(4) = -31$$



Example

Find all the relative extrema of $f(x) = \sqrt[3]{x^3 - 3x}$

$$f'(x) = \frac{x^2 - 1}{\sqrt[3]{(x^3 - 3x)^2}}$$

$$f'(x) = 0$$

$$x^2 - 1 = 0$$

or

$$x^3 - 3x = 0$$

$f'(x)$ undefined

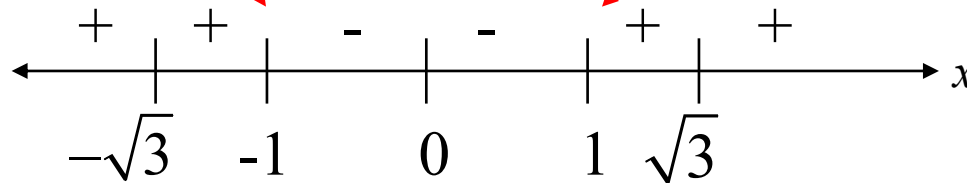
$$x = 0, \pm 1, \pm \sqrt{3}$$

Relative max.

$$f(-1) = \sqrt[3]{2}$$

Relative min.

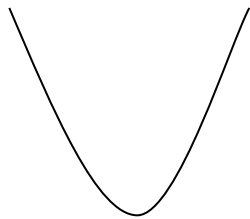
$$f(1) = -\sqrt[3]{2}$$



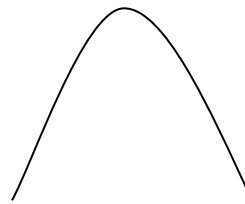
Concavity

Let f be a differentiable function on (a, b) .

1. f is *concave upward* on (a, b) if f' is increasing on (a, b) . That is, $f''(x) > 0$ for each value of x in (a, b) .
2. f is *concave downward* on (a, b) if f' is decreasing on (a, b) . That is, $f''(x) < 0$ for each value of x in (a, b) .



concave upward



concave downward

Determining the Intervals of Concavity

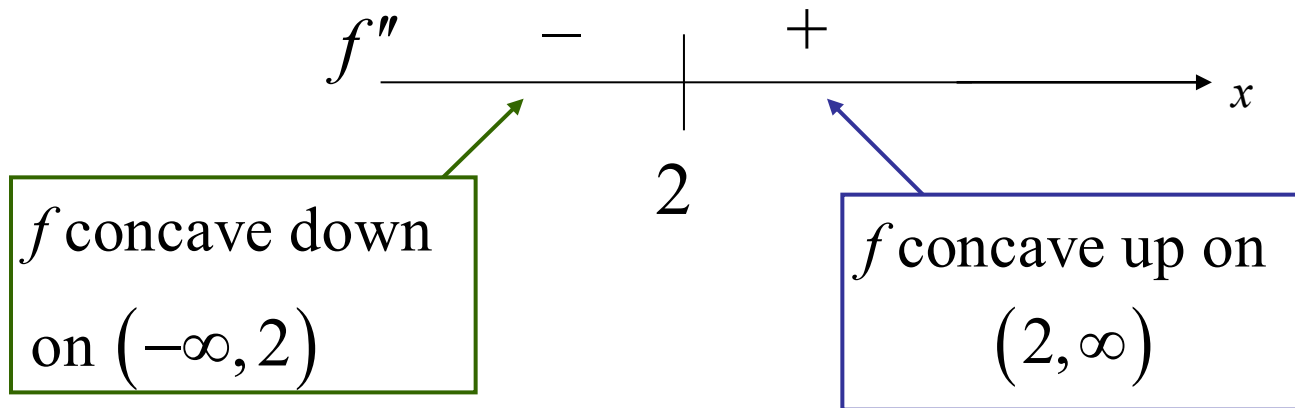
1. Determine the values for which the second derivative of f is zero or undefined. Identify the open intervals with these points.
2. Determine the sign of f'' in each interval from step 1 by testing it at a point, c , on the interval.
 - $f''(c) > 0$, f is **concave up** on that interval.
 - $f''(c) < 0$, f is **concave down** on that interval.

Example

Determine where the function $f(x) = x^3 - 6x^2 + 1$ is concave upward and concave downward.

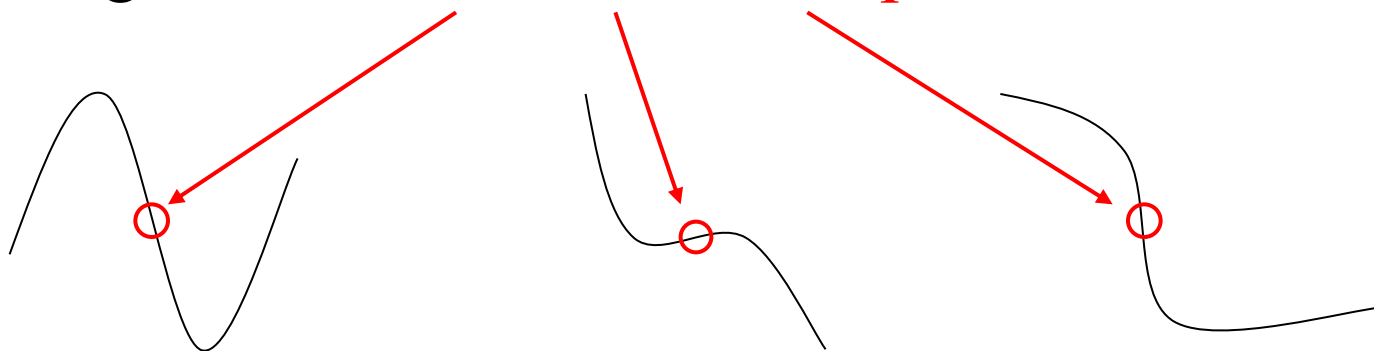
$$f'(x) = 3x^2 - 12x$$

$$f''(x) = 6x - 12 = 6(x - 2)$$



Inflection Point

A point on the graph of a continuous function f where the tangent line exists and where the concavity changes is called an inflection point.



To find inflection points, find any point, c , in the domain where $f''(x) = 0$ or $f''(x)$ is undefined.

If f'' changes sign from the left to the right of c ,
Then $(c, f(c))$ is an inflection point of f .

The Second Derivative Test

1. Compute $f'(x)$ and $f''(x)$.
2. Find all critical numbers, c , at which $f'(c) = 0$.

If	Then
$f''(c) < 0$	f has a relative maximum at c .
$f''(c) > 0$	f has a relative minimum at c .
$f''(c) = 0$	The test is inconclusive.

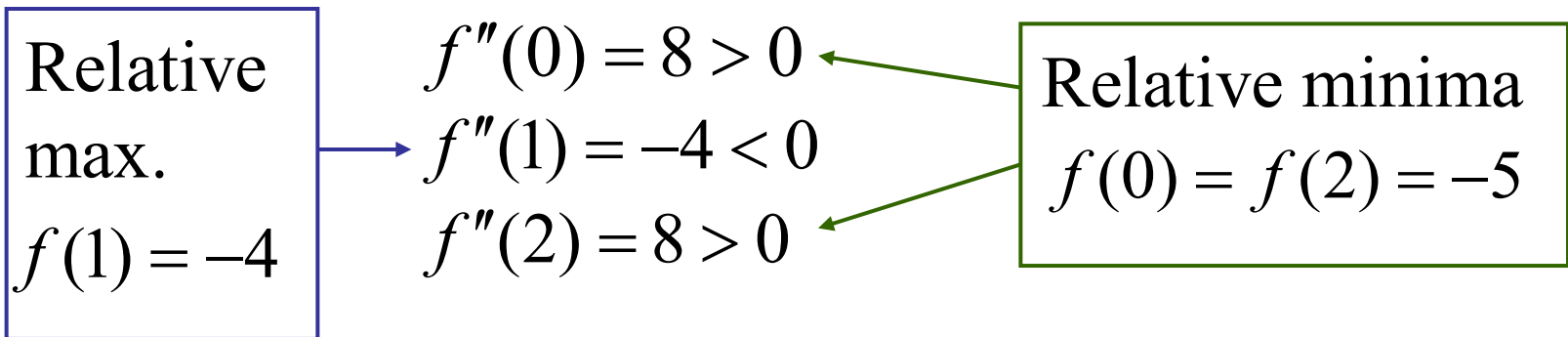
Example 1

Classify the relative extrema of $f(x) = x^4 - 4x^3 + 4x^2 - 5$ using the second derivative test.

$$\begin{aligned}f'(x) &= 4x^3 - 12x^2 + 8x \\ &= 4x(x-2)(x-1)\end{aligned}$$

Critical numbers: $x = 0, 1, 2$

$$f''(x) = 12x^2 - 24x + 8$$



Example 2

An efficiency study conducted for Elektra Electronics showed that the number of Space Commander walkie-talkies assembled by the average worker t hr after starting work at 8 A.M. is given by

$$N(t) = -t^3 + 6t^2 + 15t \quad (0 \leq t \leq 4)$$

At what time during the morning shift is the average worker performing at peak efficiency?

Step 1. Find the first and the second derivatives of $N(t)$.

$$N'(t) = -3t^2 + 12t + 15 \quad \text{and} \quad N''(t) = -6t + 12$$

.....

Example 2 (cont.)

Step 2.

Peak efficiency means that the rate of growth is maximal, that occurs at the point of inflection.

$$N''(t) = 0 \implies t = 2.$$

At 10:00 A.M. during the morning shift, the average worker is performing at peak efficiency.

Vertical Asymptote

The line $x = a$ is a *vertical asymptote* of the graph of a function f if either

$$\lim_{x \rightarrow a^-} f(x) \text{ or } \lim_{x \rightarrow a^+} f(x)$$

is infinite.

Horizontal Asymptote

The line $y = b$ is a *horizontal asymptote* of the graph of a function f if

$$\lim_{x \rightarrow -\infty} f(x) = b \text{ or } \lim_{x \rightarrow \infty} f(x) = b$$

Finding Vertical Asymptotes of Rational Functions

If $f(x) = \frac{P(x)}{Q(x)}$

is a rational function, then $x = a$ is a vertical asymptote if $Q(a) = 0$ but $P(a) \neq 0$.

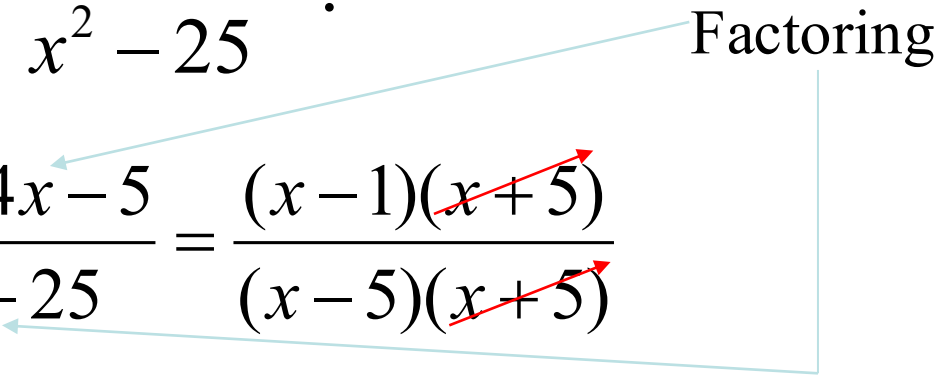
Ex. $f(x) = \frac{3x-1}{x-5}$

f has a vertical asymptote at $x = 5$.

Example

Find the vertical asymptote for the function

$$f(x) = \frac{x^2 + 4x - 5}{x^2 - 25}.$$


$$f(x) = \frac{x^2 + 4x - 5}{x^2 - 25} = \frac{(x-1)(x+5)}{(x-5)(x+5)}$$

$$f(x) = \frac{x-1}{x-5} \quad \text{when } x \neq -5$$

f has a vertical asymptote at $x = 5$.

Finding Horizontal Asymptotes of Rational Functions

Ex. $f(x) = \frac{3x^2 + 2x - 1}{x - 5x^2}$

$$\lim_{x \rightarrow \pm\infty} \frac{3x^2 + 2x - 1}{x - 5x^2} = \lim_{x \rightarrow \pm\infty} \frac{3 + \frac{2}{x} - \frac{1}{x^2}}{\frac{1}{x} - 5}$$

Divide
by x^2

f has a horizontal asymptote at $y = -\frac{3}{5}$.

Curve Sketching Guide

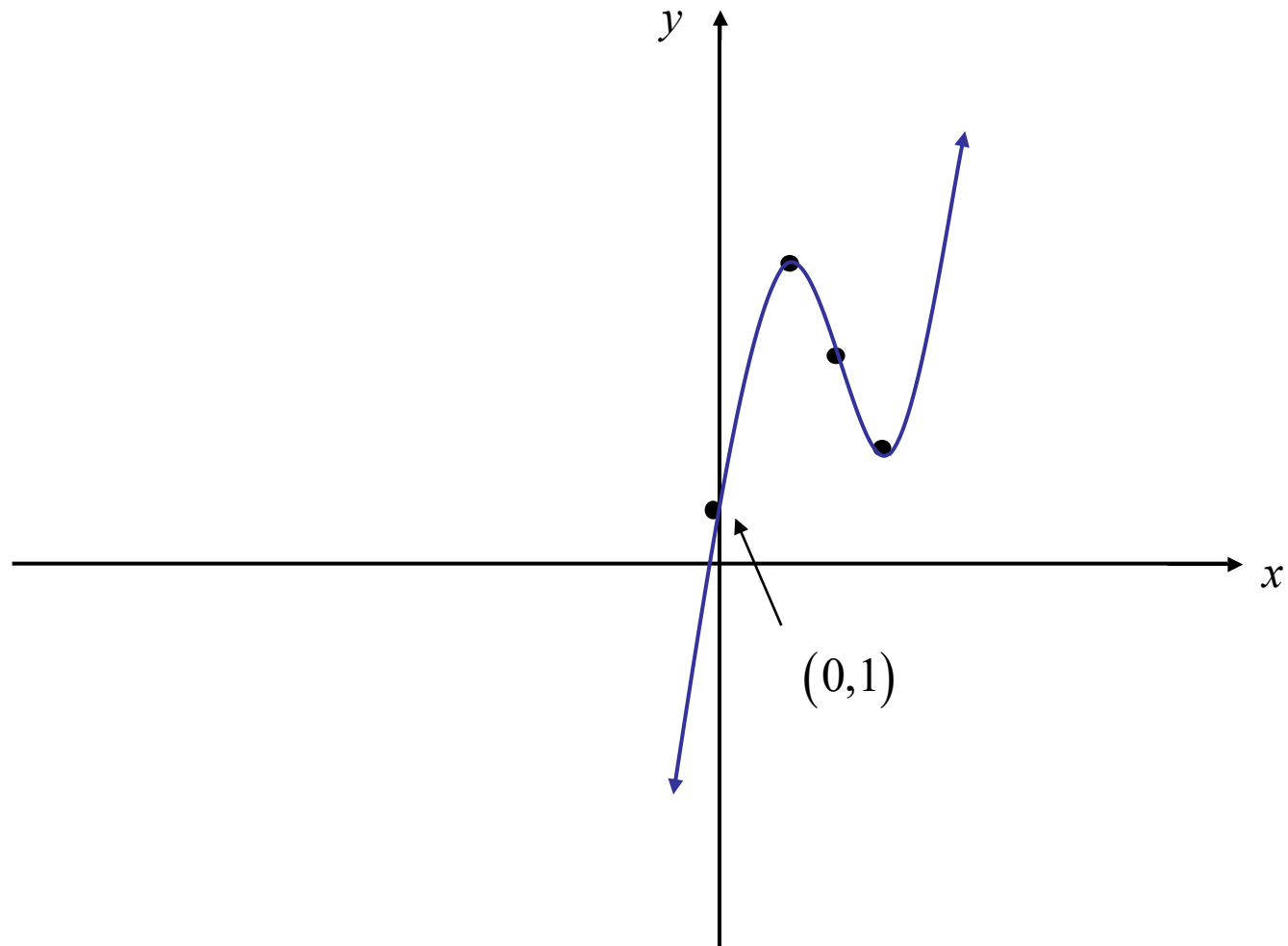
1. Determine the domain of f .
2. Find the intercepts of f if possible.
3. Look at end behavior of f .
4. Find all horizontal and vertical asymptotes.
5. Determine intervals where f is inc./dec.
6. Find the relative extrema of f .
7. Determine the concavity of f .
8. Find the inflection points of f .
9. Sketch f , use additional points as needed.

Example

Sketch: $f(x) = x^3 - 6x^2 + 9x + 1$

1. Domain: $(-\infty, \infty)$.
2. Intercept: $(0, 1)$
3. $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow -\infty} f(x) = -\infty$
4. No Asymptotes
5. $f'(x) = 3x^2 - 12x + 9$; f inc. on $(-\infty, 1) \cup (3, \infty)$; dec. on $(1, 3)$.
6. Relative max.: $(1, 5)$; relative min.: $(3, 1)$
7. $f''(x) = 6x - 12$; f concave down $(-\infty, 2)$; up on $(2, \infty)$.
8. Inflection point: $(2, 3)$

Sketch: $f(x) = x^3 - 6x^2 + 9x + 1$

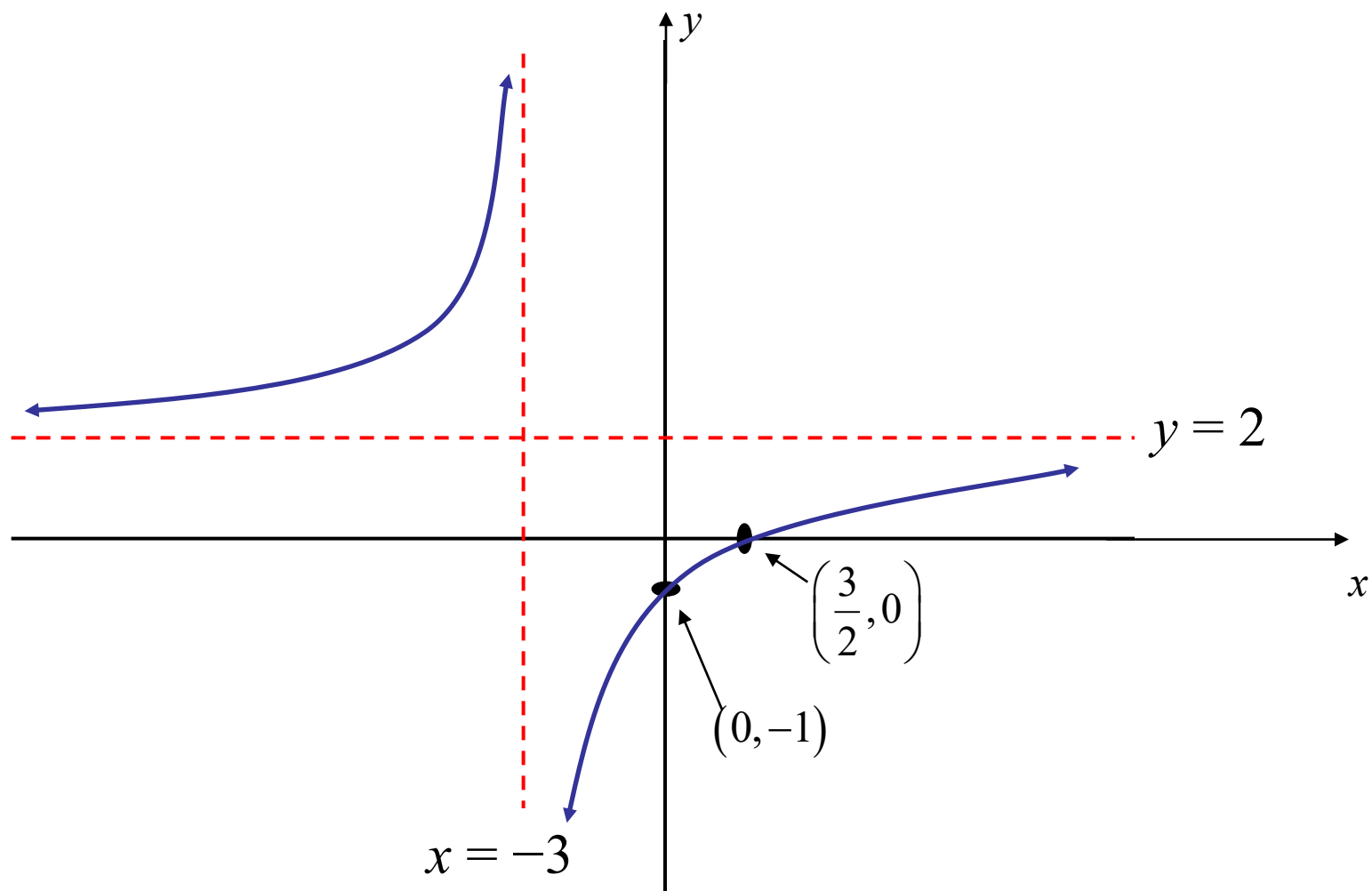


Example

Sketch: $f(x) = \frac{2x-3}{x+3}$

1. Domain: $x \neq -3$
2. Intercepts: $(0, -1)$ and $(3/2, 0)$
3. $\lim_{x \rightarrow \infty} \frac{2x-3}{x+3} = 2$ and $\lim_{x \rightarrow -\infty} \frac{2x-3}{x+3} = 2$
4. Horizontal: $y = 2$; Vertical: $x = -3$
5. $f'(x) = \frac{6}{(x+3)^2}$; f is increasing on $(-\infty, -3) \cup (-3, \infty)$.
6. No relative extrema.
7. $f''(x) = \frac{-18}{(x+3)^3}$; f is concave down on $(-3, \infty)$ and concave up on $(-\infty, -3)$.
8. No inflection points

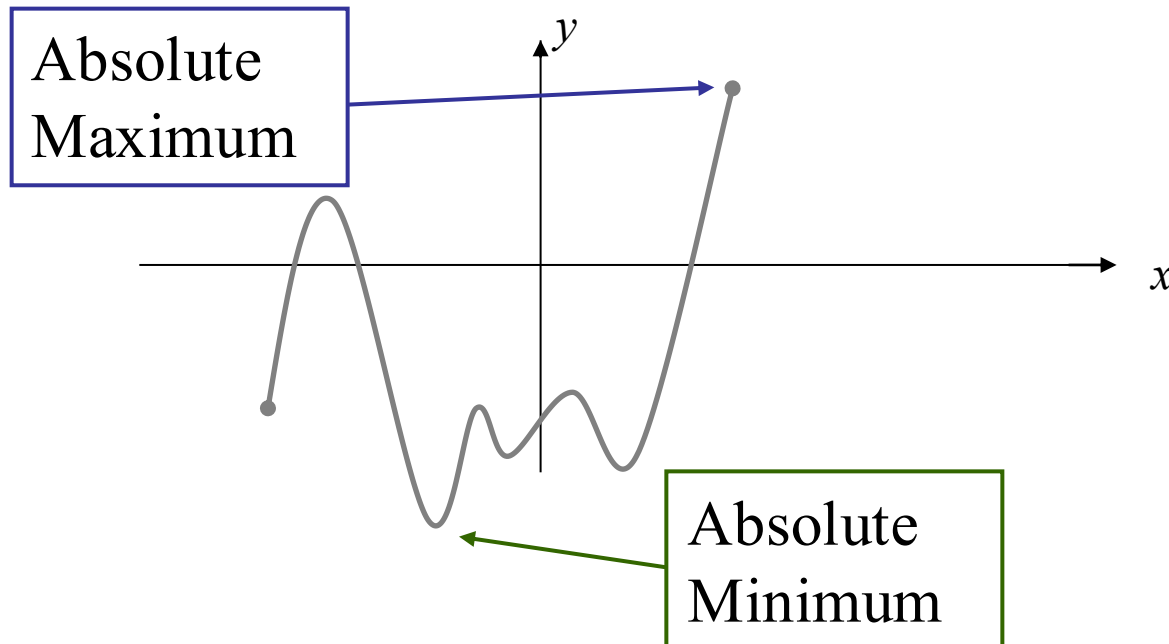
Sketch: $f(x) = \frac{2x-3}{x+3}$



Absolute Extrema

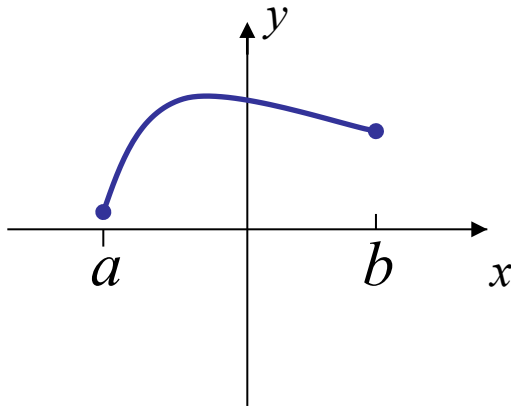
A function f has an absolute maximum at $x = c$ if $f(x) \leq f(c)$ for all x in the domain of f .

A function f has a absolute minimum at $x = c$ if $f(x) \geq f(c)$ for all x in the domain of f .

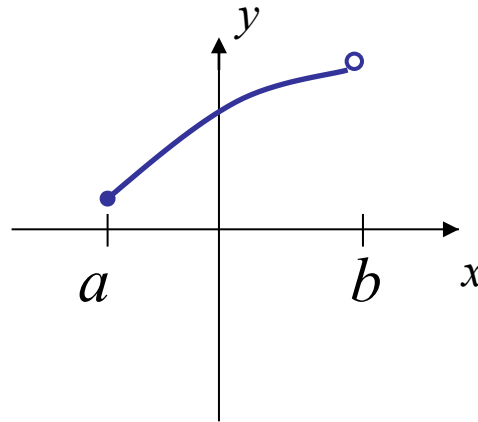


Absolute Extrema

If a function f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum and minimum on $[a, b]$.

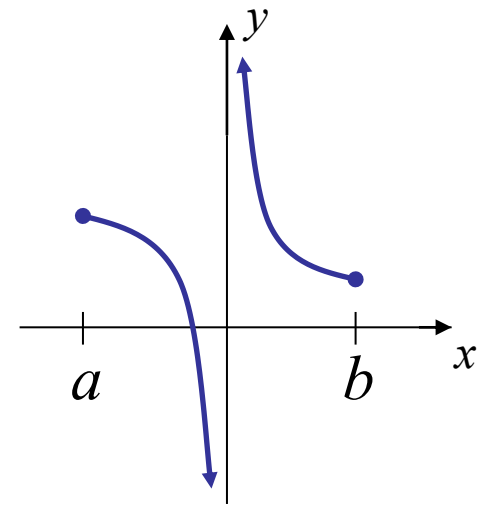


Attains max.
and min.



Attains min.
but not max.

Interval open



No min. and
no max.

Not continuous

Finding Absolute Extrema on a Closed Interval

1. Find the critical numbers of f that lie in (a, b) .
2. Compute f at each critical number as well as each endpoint.

Largest value = Absolute Maximum

Smallest value = Absolute Minimum

Example

Find the absolute extrema of $f(x) = x^3 - 3x^2$ on $\left[-\frac{1}{2}, 4\right]$.

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

Critical values at $x = 0, 2$

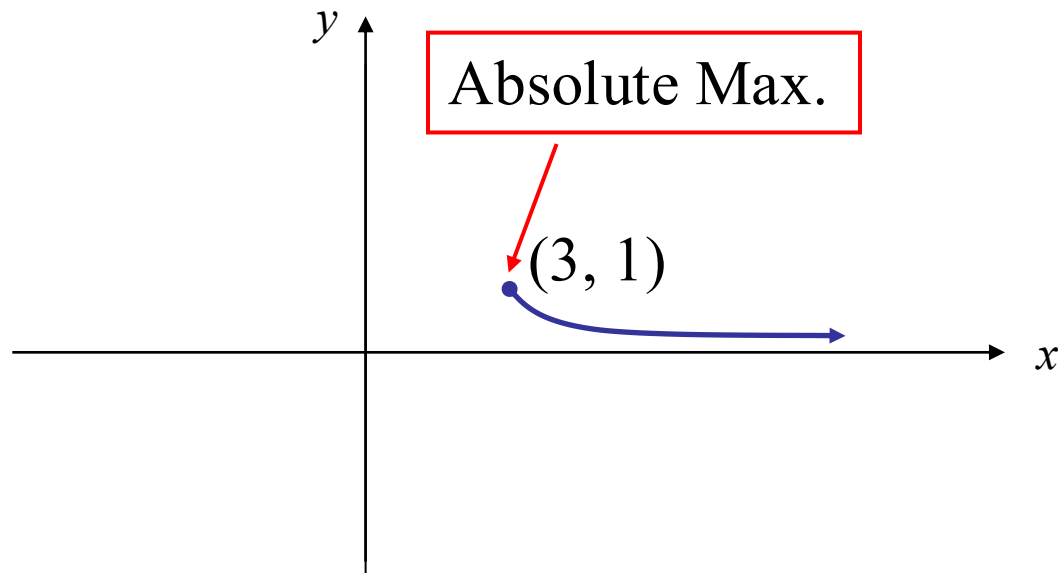
Evaluate

{	$f(0) = 0$	
	$f(2) = -4$	← Absolute Min.
	$f\left(-\frac{1}{2}\right) = -\frac{7}{8}$	
	$f(4) = 16$	← Absolute Max.

Example

Find the absolute extrema of $f(x) = \frac{1}{(x-2)}$ on $[3, \infty)$.

Notice that the interval is not closed.
Look graphically:

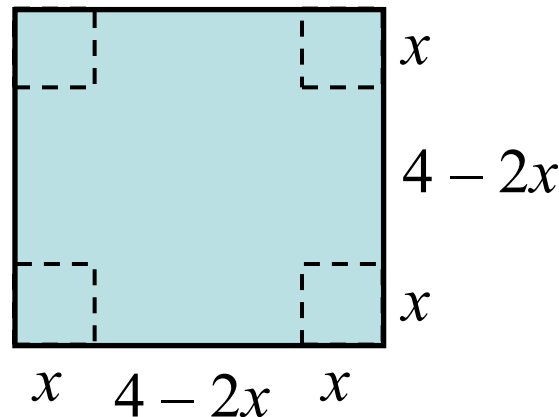


Optimization Problems

1. Assign a letter to each variable mentioned in the problem. Draw and label figure as needed.
2. Find an expression for the quantity to be optimized.
3. Use conditions to write expression as a function in one variable (note any domain restrictions).
4. Optimize the function.

Example

An open box is formed by cutting identical squares from each corner of a 4 in. by 4 in. sheet of cardboard. Find the dimensions of the box that will yield the maximum volume.



$$V = lwh = (4 - 2x)(4 - 2x)x; x \text{ in } [0, 2] \quad \dots\dots$$

$$V(x) = 16x - 16x^2 + 4x^3$$

$$\begin{aligned} V'(x) &= 16 - 32x + 12x^2 \\ &= 4(2 - 3x)(2 - x) \end{aligned}$$

Critical points: $x = 2, \frac{2}{3}$

$$V(2) = 0$$

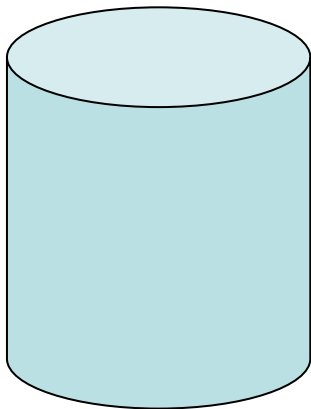
$$V(0) = 0$$

$$V\left(\frac{2}{3}\right) \approx 4.74 \text{ in}^3$$

The dimensions are $\frac{8}{3}$ in. by $\frac{8}{3}$ in. by $\frac{2}{3}$ in.
giving a maximum box volume of 4.74 in^3 .

Example

An metal can with volume 60 in^3 is to be constructed in the shape of a right circular cylinder. If the cost of the material for the side is $\$0.05/\text{in.}^2$ and the cost of the material for the top and bottom is $\$0.03/\text{in.}^2$ Find the dimensions of the can that will minimize the cost.



$$V = \pi r^2 h = 60$$

$$C = (0.03)(2)\pi r^2 + (0.05)2\pi r h$$

cost

top and
bottom

side

.....

$$V = \pi r^2 h = 60 \quad \text{So } h = \frac{60}{\pi r^2}$$

$$C = (0.03)(2)\pi r^2 + (0.05)2\pi r h$$

$$= (0.03)(2)\pi r^2 + (0.05)2\pi r \frac{60}{\pi r^2}$$

Sub. in for h

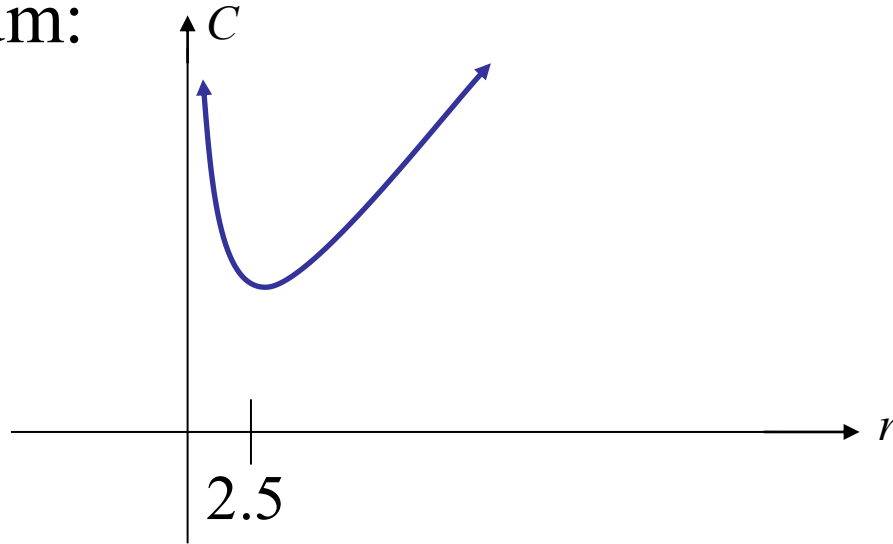
$$= 0.06\pi r^2 + \frac{6}{r}$$

$$C' = 0.12\pi r - \frac{6}{r^2}$$

$$C' = 0 \quad \text{gives} \quad 0.12\pi r = \frac{6}{r^2}$$

$$r = \sqrt[3]{\frac{6}{0.12\pi}} \approx 2.52 \text{ in. which yields } h \approx 3.02 \text{ in.}$$

Graph of cost function to verify absolute minimum:



So with a radius ≈ 2.52 in. and height ≈ 3.02 in., the cost is minimized at $\approx \$3.58$.

Common Steps to Solve Optimization Problems:

1. **Define Variables:** Identify the variables and constants, and create a sketch if necessary.
2. **Define the Objective Function:** Formulate the equation representing the quantity to be optimized.
3. **Define Constraints:** Find the equations that restrict the variables.
4. **Express as a Single Variable:** Substitute the constraint into the objective function to get a function $f(x)$.
5. **Differentiate and Solve:** Find the derivative $f'(x)$, set it to zero, and solve for critical points.

Applications of Optimization

Area Maximization: Finding dimensions of a rectangular field to maximize area given a fixed amount of fencing, such as a 2,400 ft fence creating a 600 ft x 1,200 ft area.

Cost Minimization: Designing a container with a specific volume (e.g., an aquarium) while minimizing the surface area of glass used.

Product Maximization: Identifying two positive numbers whose product is 750, while minimizing the sum of one and ten times the other.

Volume Maximization: Creating an open-top box with the largest possible volume by cutting squares from the corners of a sheet of cardboard.

Distance/Time Minimization: Finding the shortest path for a pipeline or travel route, such as minimizing the cost of running a cable from a power station to an island.

Profit Optimization: Determining the number of items a company should produce to maximize profit while minimizing production and labor costs.

Calculus is used in **optimizing machine learning models** by finding the minimum or maximum values of a function, which helps in improving the accuracy and efficiency of the model. This is achieved by using techniques such as gradient descent and optimization algorithms to adjust the model's parameters.

Gradient Vectors of Two Functions in Multivariable Calculus

$$\nabla f_1(x, y) = \frac{\partial f_1}{\partial x} \mathbf{i} + \frac{\partial f_1}{\partial y} \mathbf{j} = \mathbf{i} + \mathbf{j}$$

$$\nabla f_2(x, y) = \frac{\partial f_2}{\partial x} \mathbf{i} + \frac{\partial f_2}{\partial y} \mathbf{j} = 2x\mathbf{i} + 2y\mathbf{j}$$

